XDJ-X

All-in-one DJ system

Exchange protocol

Service commands

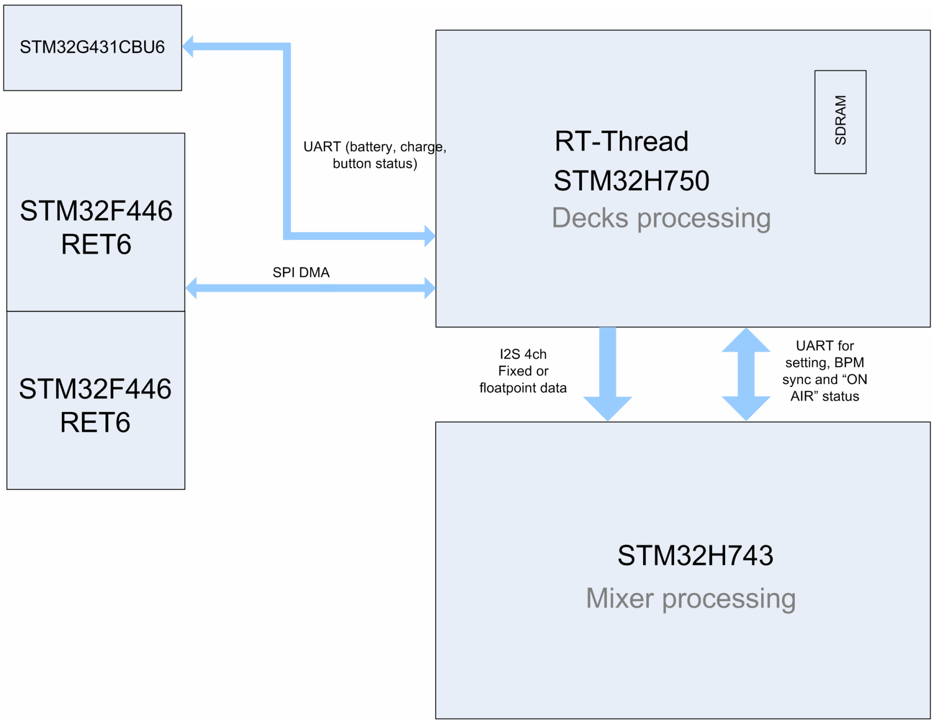
Assembly name:	**Firmware version:**
Display module	0.83
Mixer assy	0.35
Deck assy	0.29
Power manager	0.25

Last modified date:

30.04.2025

Content

General connection diagram exchange date:



Display module <=> Power manager

UART transfer 115200 baud.

Power manager => Display module:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bytes	0xAX								0xXX							
Bits	1	0	1	0	0	0	X	X	X	X	X	X	X	X	X	X

Byte[0] - status:

A1: no AC

A2: AC charge

A3: full battery

Byte[1] - data:

0...100: charge precentage (when the value changes or on request)

0xFE: button pressed

0xFD: button unpressed

0xFC: shutdown command

Display module => Power manager:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bytes	0xA1								0xXX							
Bits	1	0	1	0	0	0	0	1	X	X	X	X	X	X	X	X

Byte[0] = 0xA1

Byte[1]:

0x55: battery percentage request

0xFC: shutdown permission

Display module <=> Deck assy

SPI DMA transfer

Deck assy => Display module:

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
Bytes	PADS								BUTTONS							
Bits	PD7	PD6	PD5	PD4	PD3	PD2	PD1	PD0	RLP	LP8	LP4	VNL	SLIP	REV	CUE	PLAY

	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
Bytes	BUTTONS						ADC_PITCH (10 bits)									
Bits	JSNS	TMP	NXT	PREV	RD	0	P	P	P	P	P	P	P	P	P	P

	20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F
Bytes									JOG_SPEED_MSB							
Bits	DIR	FW	FW	FW	FW	FW	FW	FW	X	X	X	X	X	X	X	X

	30	31	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F
Bytes	JOG_SPEED_LSB								CRC							
Bits	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

ADC_PITCH 0…1023, 512 – center zone

RD – rotation detect

DIR - jog wheel rotation direction

FW – firmware version in code (0-127)

JOG_SPEED 0…65535 16 bits resolution (when jog stopped speed = 0xFFFF)

CRC = (0byte + 1byte +…+6byte)%256

Display module => Deck assy:

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
Bytes	NUM_P								PADn_RED							
Bits	RNGr	RNGg	RNGb	LD1	LD0	nP	nP	nP	R	R	R	R	R	R	R	R

	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
Bytes	PADn_GREEN								PADn_BLUE							
Bits	G	G	G	G	G	G	G	G	B	B	B	B	B	B	B	B

	20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F
Bytes	PLAY_POS								SLIP_POS							
Bits	X	X	X	X	X	X	X	X	0	X	X	X	X	X	X	X

	30	31	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F

Bytes	CUE_POS								CRC							
Bits	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

nP	LD1	LD0
0	CUE	PLAY
1	SLIP	REV
2	LP4	VNL
3	RLP	LP8
4	TMP	VNLr
5	SNSr	MEMr
6	0	0
7	0	0

nP - package number and at the same time pad number
PADn_(RED, GREEN, BLUE) RGB 888 color for PAD[NUM_P]
RNGr, RNGg, RNGb – RGB bits <<[NUM_P]
MEMr, SNSr, VNLr – VFD segments
SLIP_POS 0…84. 85 – SLIP disable
CUE_POS 0…84. 85 – CUE without cue on jog display
PLAY_POS 0…134 – jog display position cursor 139 = power on animation 138 = eject animation 137 = load in animation 136 = fill circle on display 135 = empty circle.
CRC = (0byte + 1byte +…+6byte)%256

Display module <=> Mixer assy

UART transfer 115200 baud.

Display module => Mixer assy:

Mixer assy => Display module:
ON AIR